



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°4_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMES
INFORMATIQUES

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L'objectif de ce TD est de:

1. analyser les ports ouverts
2. configurer un firewall simple
3. écrire des règles de filtrage réseau
4. comprendre les chaînes de filtrage
5. comparer différentes solutions de firewall Linux
6. analyser une configuration SSH
7. sécuriser les accès
8. mettre en place l'authentification par clé
9. limiter les connexions SSH
10. appliquer les bonnes pratiques de sécurité

1. Analyse du serveur

Identification des ports ouverts

```

lens@lens:~$ ss -tuln
Netid State  Recv-Q  Send-Q           Local Address:Port      Peer Address:Port
Process
udp    UNCONN  0        0           127.0.0.54:53            0.0.0.0:*
udp    UNCONN  0        0           127.0.0.53%lo:53         0.0.0.0:*
udp    UNCONN  0        0       10.79.242.79%enp0s3:68    0.0.0.0:*
udp    UNCONN  0        0           0.0.0.0:111              0.0.0.0:*
udp    UNCONN  0        0           127.0.0.1:161            0.0.0.0:*
udp    UNCONN  0        0           [::]:111                 [::]:*
udp    UNCONN  0        0           [::1]:161                [::]:*
tcp    LISTEN  0        4096           0.0.0.0:10050            0.0.0.0:*
tcp    LISTEN  0        4096       127.0.0.53%lo:53         0.0.0.0:*
tcp    LISTEN  0        4096           0.0.0.0:22               0.0.0.0:*
tcp    LISTEN  0        100           0.0.0.0:25               0.0.0.0:*
tcp    LISTEN  0        511           0.0.0.0:80               0.0.0.0:*
tcp    LISTEN  0        4096           0.0.0.0:111              0.0.0.0:*
tcp    LISTEN  0        4096       127.0.0.54:53            0.0.0.0:*
tcp    LISTEN  0        4096           [::]:10050               [::]:*
tcp    LISTEN  0        4096           [::]:22                  [::]:*
tcp    LISTEN  0        100           [::]:25                  [::]:*
tcp    LISTEN  0        4096           [::]:111                 [::]:*

```

Vérification des services actifs

```
lens@lens: ~  
lens@lens:~$ systemctl list-units --type=service  
UNIT  
● apache2.service  
  apparmor.service  
  apport.service  
  blk-availability.service  
  console-setup.service  
  cron.service  
  dbus.service  
  finalrd.service  
  getty@tty1.service  
  keyboard-setup.service  
  kmod-static-nodes.service  
  lvm2-monitor.service  
  ModemManager.service  
  multipathd.service  
  networkd-dispatcher.service  
  nginx.service  
  php8.3-fpm.service  
  plymouth-quit-wait.service  
  plymouth-quit.service  
  plymouth-read-write.service  
  polkit.service  
  postfix.service  
  postfix@-.service  
  rpcbind.service  
  rsyslog.service  
  setvtrgb.service  
  snapd.apparmor.service  
  snapd.seeded.service  
  snmpd.service  
  ssh.service  
  sysstat.service  
  systemd-binfmt.service  
  systemd-fsck@dev-disk-by\x2duuid-ddf25e8d\x2d86fd\x2d4788\x2d9212\x2d8c82>  
  systemd-journal-flush.service  
  systemd-journald.service  
  systemd-logind.service  
  systemd-modules-load.service  
  systemd-networkd-wait-online.service
```

2. Firewall simple avec UFW

Installer UFW, politique par défaut, autorisation des services nécessaires et activation du firewall

```

lens@lens:~$ sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
lens@lens:~$ sudo ufw status
Status: inactive
lens@lens:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
lens@lens:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
lens@lens:~$ sudo ufw allow 22
Rules updated
Rules updated (v6)
lens@lens:~$ sudo ufw allow 80
Rules updated
Rules updated (v6)
lens@lens:~$ sudo ufw allow 443
Rules updated
Rules updated (v6)
lens@lens:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
lens@lens:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22 ALLOW IN Anywhere
80 ALLOW IN Anywhere
443 ALLOW IN Anywhere
22 (v6) ALLOW IN Anywhere (v6)
80 (v6) ALLOW IN Anywhere (v6)
443 (v6) ALLOW IN Anywhere (v6)

```

3. Configuration avec iptables

Afficher les règles actuelles, blocage de tout le trafic entrant, autorisation SSH, de HTTP, de HTTPS et autoriser les connexions déjà établies.

```

lens@lens: ~
lens@lens:~$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source                destination
ufw-before-logging-input all -- anywhere             anywhere
ufw-before-input all -- anywhere             anywhere
ufw-after-input all -- anywhere             anywhere
ufw-after-logging-input all -- anywhere             anywhere
ufw-reject-input all -- anywhere             anywhere
ufw-track-input all -- anywhere             anywhere
ACCEPT     tcp -- anywhere             anywhere             tcp dpt:ssh

Chain FORWARD (policy DROP)
target     prot opt source                destination
ufw-before-logging-forward all -- anywhere             anywhere
ufw-before-forward all -- anywhere             anywhere
ufw-after-forward all -- anywhere             anywhere
ufw-after-logging-forward all -- anywhere             anywhere
ufw-reject-forward all -- anywhere             anywhere
ufw-track-forward all -- anywhere             anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
ufw-before-logging-output all -- anywhere             anywhere
ufw-before-output all -- anywhere             anywhere
ufw-after-output all -- anywhere             anywhere
ufw-after-logging-output all -- anywhere             anywhere
ufw-reject-output all -- anywhere             anywhere
ufw-track-output all -- anywhere             anywhere

Chain ufw-after-forward (1 references)
target     prot opt source                destination

Chain ufw-after-input (1 references)
target     prot opt source                destination
ufw-skip-to-policy-input udp -- anywhere             anywhere             udp dpt:netbios-ns
ufw-skip-to-policy-input udp -- anywhere             anywhere             udp dpt:netbios-dgm
ufw-skip-to-policy-input tcp -- anywhere             anywhere             tcp dpt:netbios-ssn
ufw-skip-to-policy-input tcp -- anywhere             anywhere             tcp dpt:microsoft-ds
ufw-skip-to-policy-input udp -- anywhere             anywhere             udp dpt:bootps
ufw-skip-to-policy-input udp -- anywhere             anywhere             udp dpt:bootpc
ufw-skip-to-policy-input all -- anywhere             anywhere             ADDRTYPE match dst-type BROADCAST

```

```

Chain ufw-after-logging-forward (1 references)
target    prot opt source                destination
LOG       all  -- anywhere             anywhere
limit: avg 3/min burst 10 LOG level warn prefix "[UFW BLOCK] "

Chain ufw-after-logging-input (1 references)
target    prot opt source                destination
LOG       all  -- anywhere             anywhere
limit: avg 3/min burst 10 LOG level warn prefix "[UFW BLOCK] "

Chain ufw-after-logging-output (1 references)
target    prot opt source                destination

Chain ufw-after-output (1 references)
target    prot opt source                destination

Chain ufw-before-forward (1 references)
target    prot opt source                destination
ACCEPT    all  -- anywhere             anywhere          ctstate RELATED,ESTABLISHED
ACCEPT    icmp -- anywhere             anywhere          icmp destination-unreachable
ACCEPT    icmp -- anywhere             anywhere          icmp time-exceeded
ACCEPT    icmp -- anywhere             anywhere          icmp parameter-problem
ACCEPT    icmp -- anywhere             anywhere          icmp echo-request
ufw-user-forward all -- anywhere             anywhere

Chain ufw-before-input (1 references)
target    prot opt source                destination
ACCEPT    all  -- anywhere             anywhere
ACCEPT    all  -- anywhere             anywhere          ctstate RELATED,ESTABLISHED
ufw-logging-deny all -- anywhere             anywhere          ctstate INVALID
DROP      all  -- anywhere             anywhere          ctstate INVALID
ACCEPT    icmp -- anywhere             anywhere          icmp destination-unreachable
ACCEPT    icmp -- anywhere             anywhere          icmp time-exceeded
ACCEPT    icmp -- anywhere             anywhere          icmp parameter-problem
ACCEPT    icmp -- anywhere             anywhere          icmp echo-request
ACCEPT    udp  -- anywhere             anywhere          udp spt:bootps dpt:bootpc
ufw-not-local all -- anywhere             anywhere
ACCEPT    udp  -- anywhere             mdns.mcast.net    udp dpt:mdns
ACCEPT    udp  -- anywhere             239.255.255.250   udp dpt:1900
ufw-user-input all -- anywhere             anywhere

Chain ufw-before-logging-forward (1 references)

```

```

lens@lens: ~
ufw-user-input all -- anywhere             anywhere

Chain ufw-before-logging-forward (1 references)
target    prot opt source                destination

Chain ufw-before-logging-input (1 references)
target    prot opt source                destination

Chain ufw-before-logging-output (1 references)
target    prot opt source                destination

Chain ufw-before-output (1 references)
target    prot opt source                destination
ACCEPT    all  -- anywhere             anywhere
ACCEPT    all  -- anywhere             anywhere          ctstate RELATED,ESTABLISHED
ufw-user-output all -- anywhere             anywhere

Chain ufw-logging-allow (0 references)
target    prot opt source                destination
LOG       all  -- anywhere             anywhere
limit: avg 3/min burst 10 LOG level warn prefix "[UFW ALLOW] "

Chain ufw-logging-deny (2 references)
target    prot opt source                destination
RETURN    all  -- anywhere             anywhere          ctstate INVALID limit: avg 3/min burst 10
LOG       all  -- anywhere             anywhere          limit: avg 3/min burst 10 LOG level warn prefix "[UFW BLOCK] "

Chain ufw-not-local (1 references)
target    prot opt source                destination
RETURN    all  -- anywhere             anywhere          ADDRTYPE match dst-type LOCAL
RETURN    all  -- anywhere             anywhere          ADDRTYPE match dst-type MULTICAST
RETURN    all  -- anywhere             anywhere          ADDRTYPE match dst-type BROADCAST
ufw-logging-deny all -- anywhere             anywhere          limit: avg 3/min burst 10
DROP      all  -- anywhere             anywhere

Chain ufw-reject-forward (1 references)
target    prot opt source                destination

Chain ufw-reject-input (1 references)
target    prot opt source                destination

Chain ufw-reject-output (1 references)

```

```

lens@lens: ~
Chain ufw-reject-output (1 references)
target      prot opt source      destination

Chain ufw-skip-to-policy-forward (0 references)
target      prot opt source      destination
DROP        all  --  anywhere    anywhere

Chain ufw-skip-to-policy-input (7 references)
target      prot opt source      destination
DROP        all  --  anywhere    anywhere

Chain ufw-skip-to-policy-output (0 references)
target      prot opt source      destination
ACCEPT      all  --  anywhere    anywhere

Chain ufw-track-forward (1 references)
target      prot opt source      destination

Chain ufw-track-input (1 references)
target      prot opt source      destination

Chain ufw-track-output (1 references)
target      prot opt source      destination
ACCEPT      tcp  --  anywhere    anywhere    ctstate NEW
ACCEPT      udp  --  anywhere    anywhere    ctstate NEW

Chain ufw-user-forward (1 references)
target      prot opt source      destination

Chain ufw-user-input (1 references)
target      prot opt source      destination
ACCEPT      tcp  --  anywhere    anywhere    tcp dpt:ssh
ACCEPT      udp  --  anywhere    anywhere    udp dpt:22
ACCEPT      tcp  --  anywhere    anywhere    tcp dpt:http
ACCEPT      udp  --  anywhere    anywhere    udp dpt:80
ACCEPT      tcp  --  anywhere    anywhere    tcp dpt:https
ACCEPT      udp  --  anywhere    anywhere    udp dpt:https

Chain ufw-user-limit (0 references)
target      prot opt source      destination
LOG         all  --  anywhere    anywhere    limit: avg 3/min burst 5 LOG level warn prefix "[UFW LIMIT BLOCK] "

Chain ufw-user-limit-accept (0 references)
target      prot opt source      destination
ACCEPT      all  --  anywhere    anywhere

Chain ufw-user-logging-forward (0 references)
target      prot opt source      destination

Chain ufw-user-logging-input (0 references)
target      prot opt source      destination

Chain ufw-user-logging-output (0 references)
target      prot opt source      destination

Chain ufw-user-output (1 references)
target      prot opt source      destination
lens@lens:~$ sudo iptables -P INPUT DROP
lens@lens:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
lens@lens:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
lens@lens:~$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
lens@lens:~$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
lens@lens:~$ |

```

4. Firewall moderne avec nftables

```
lens@lens:~$ sudo apt install nftables -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nftables is already the newest version (1.0.9-1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
lens@lens:~$ sudo systemctl enable nftables
lens@lens:~$ sudo nano /etc/nftables.conf
lens@lens:~$ sudo systemctl restart nftables
lens@lens:~$ |
```

```
lens@lens: ~
GNU nano 7.2 /etc/nftables.conf
#!/usr/sbin/nft -f

flush ruleset

table inet filter {
chain input {
type filter hook input priority 0;
policy drop;
ct state established,related accept
tcp dport 22 accept
tcp dport 80 accept
tcp dport 443 accept
}
}

[ Read 14 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

5. Tests de sécurité

```
lens@lens:~$ nmap localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-24 21:37 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00093s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 4.76 seconds
lens@lens:~$
```

6. Analyse de la configuration actuelle

```
lens@lens:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: ➤
   Active: active (running) since Tue 2026-03-24 20:43:14 UTC; 1h 2min ago
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 875 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 919 (sshd)
     Tasks: 1 (limit: 2263)
    Memory: 4.1M (peak: 5.4M)
       CPU: 180ms
    CGroup: /system.slice/ssh.service
            └─919 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 24 20:43:12 lens systemd[1]: Starting ssh.service - OpenBSD Secure Shell
Mar 24 20:43:14 lens sshd[919]: Server listening on 0.0.0.0 port 22.
Mar 24 20:43:14 lens sshd[919]: Server listening on :: port 22.
Mar 24 20:43:14 lens systemd[1]: Started ssh.service - OpenBSD Secure Shell
Mar 24 20:53:56 lens sshd[1561]: Accepted password for lens from 192.168.43
Mar 24 20:53:56 lens sshd[1561]: pam_unix(sshd:session): session opened for
lines 1-20/20 (END)
```

```
lens@lens: ~$ sudo cat /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:
```

- Le port utilisé est le port 22.

```
# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
```

- **PermitRootLogin yes, donc root autorisé**

```
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no
```

- **PubkeyAuthentication yes, donc connexion par mot de passe autorisée**

7. Désactiver la connexion root

```
lens@lens:~$ sudo nano /etc/ssh/sshd_config
lens@lens:~$ sudo systemctl restart ssh
```

```

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

```

8. Authentification par clé SSH

```

lens@lens:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/lens/.ssh/id_ed25519):
/home/lens/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/lens/.ssh/id_ed25519
Your public key has been saved in /home/lens/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:kXd0nT/YQC3hYy5Fbk3qQahUpoGGlvBwM/mmP4+RF8Q lens@lens
The key's randomart image is:
+--[ED25519 256]--+
|  o.++ ...o+=+o.|
|  +=oo.o+o=++o.|
|  ..o +Eo .X=..|
|    o.+ *.oo.|
|    o S. . o .|
|    . . . .|
|    .o .|
|    o+|
|    .o.|
+-----[SHA256]-----+
lens@lens:~$ |

```

```
ineto 1e00::a00:27ff:fe17:0831/64 scope link
    valid_lft forever preferred_lft forever
lens@lens:~$ ssh-copy-id lens@192.168.43.79
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/lens/.ssh/id_ed25519.pub"
The authenticity of host '192.168.43.79 (192.168.43.79)' can't be established.
ED25519 key fingerprint is SHA256:n0DC/5aoIY4E9BhwtIzMjRr6IoPB7ncus2ADLZNPqBw.
This host key is known by the following other names/addresses:
+ ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
lens@192.168.43.79's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'lens@192.168.43.79'"
and check to make sure that only the key(s) you wanted were added.
```

```
lens@lens: ~  
lens@lens:~$ exit  
logout  
Connection to 192.168.43.79 closed.  
  
C:\Users\Lens Sandro Petiote>ssh lens@192.168.43.79  
lens@192.168.43.79's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-106-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Tue Mar 24 10:39:12 PM UTC 2026  
  
System load:  0.01          Processes:            135  
Usage of /:   48.7% of 11.21GB Users logged in:        1  
Memory usage: 13%          IPv4 address for enp0s3: 192.168.43.79  
Swap usage:   0%  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
just raised the bar for easy, resilient and secure K8s cluster deployment  
  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
Expanded Security Maintenance for Applications is not enabled.  
  
14 updates can be applied immediately.  
13 of these updates are standard security updates.  
To see these additional updates run: apt list --upgradable  
  
1 additional security update can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings  
  
Last login: Tue Mar 24 22:04:12 2026 from 192.168.43.160  
lens@lens:~$ |
```

9. Désactiver l'authentification par mot de passe

```
#      X11Forwarding no
#      AllowTcpForwarding no
#      PermitTTY no
#      ForceCommand cvs server
lens@lens:~$ sudo nano /etc/ssh/sshd_config
lens@lens:~$ sudo systemctl restart ssh
lens@lens:~$ |
```

```
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication no
#PermitEmptyPasswords no

# Global settings for the sshd daemon
```

10. Changer le port SSH

```
#      systemctl daemon-reload
#      systemctl restart ssh.socket
#
Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO
```

```
# X11Forwarding no
# AllowTcpForwarding no
# PermitTTY no
# ForceCommand cvs server
lens@lens:~$ sudo nano /etc/ssh/sshd_config
lens@lens:~$ sudo systemctl restart ssh
lens@lens:~$ |

lens@lens:~$ sudo ss -tnpt | grep ssh
lens@lens:~$ sudo ss -lnpt | grep ssh
LISTEN 0      4096      0.0.0.0:2222      0.0.0.0:*      users:((("sshd",pid=869,fd=3),("systemd",pid=1,fd=104))

LISTEN 0      4096      [::]:2222      [::]:*      users:((("sshd",pid=869,fd=4),("systemd",pid=1,fd=105))

lens@lens:~$ _

PS C:\Users\Lens Sandro Petiote> ssh lens@192.168.229 -p 2222
ssh: connect to host 192.168.0.229 port 2222: Connection timed out
PS C:\Users\Lens Sandro Petiote> |
```

- Malgré SSH écoute bien bien le port 2222, je n'arrive pas a connecter a mon server depuis mon PC windows via le port 2222

11. Protection contre brute force

```

"1490349000 test failed.dns.ch", "\s*test <F-ID>\S+</F-ID>"
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex/testcase.py:435: SyntaxWarning: invalid escape sequence '\S'
'^'+prefix+'<F-ID>User <F-USER>\S+</F-USER></F-ID> not allowed\n'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex/testcase.py:443: SyntaxWarning: invalid escape sequence '\S'
'^'+prefix+'User <F-USER>\S+</F-USER> not allowed\n'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex/testcase.py:444: SyntaxWarning: invalid escape sequence '\d'
'^'+prefix+'Received disconnect from <F-ID><ADDR> port \d+</F-ID>'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex/testcase.py:451: SyntaxWarning: invalid escape sequence '\s'
_test_variants('common', prefix="\s*\S+ sshd\[<F-MLFID>\d+</F-MLFID>\]:\s+")
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex/testcase.py:537: SyntaxWarning: invalid escape sequence '\['
'common[prefregex="\[<F-MLFID>\d+</F-MLFID>\] connect <F-CONTENT>.+</F-CONTENT>$"]'
/usr/lib/python3/dist-packages/fail2ban/tests/server/testcase.py:1375: SyntaxWarning: invalid escape sequence '\s'
'''{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-nft-mp\s+.*\s+\\Khandle\s+(\d+)\$'; } | while r
/usr/lib/python3/dist-packages/fail2ban/tests/server/testcase.py:1378: SyntaxWarning: invalid escape sequence '\s'
'''{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-mp\s+.*\s+\\Khandle\s+(\d+)\$'; } | while
/usr/lib/python3/dist-packages/fail2ban/tests/server/testcase.py:1421: SyntaxWarning: invalid escape sequence '\s'
'''{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-nft-ap\s+.*\s+\\Khandle\s+(\d+)\$'; } | while r
/usr/lib/python3/dist-packages/fail2ban/tests/server/testcase.py:1424: SyntaxWarning: invalid escape sequence '\s'
'''{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-ap\s+.*\s+\\Khandle\s+(\d+)\$'; } | while
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
Setting up python3-pyinotify (0.9.6-2ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
lens@lens:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-03-29 22:17:37 UTC; 19s ago
     Docs: man:fail2ban(1)
   Main PID: 1832 (fail2ban-server)
    Tasks: 5 (limit: 2263)
   Memory: 40.2M (peak: 40.5M)
      CPU: 541ms
   CGroup: /system.slice/fail2ban.service
           └─1832 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Mar 29 22:17:37 lens systemd[1]: Started fail2ban.service - Fail2Ban Service.
Mar 29 22:17:38 lens fail2ban-server[1832]: 2026-03-29 22:17:38,054 fail2ban.configreader [1832]: WARNING 'allowipv6'
Mar 29 22:17:38 lens fail2ban-server[1832]: Server ready
lines 1-14/14 (END)

```

```

[31] + Stopped
lens@lens:~$ sudo systemctl status fail2ban
lens@lens:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
  - Jail list:  sshd
lens@lens:~$

```

12. Tests de sécurité

```

lens@lens:~$ ss -tuln
Netid      State      Recv-Q      Send-Q      Local Address:Port
udp        UNCONN     0            0            127.0.0.54:53
udp        UNCONN     0            0            127.0.0.53%lo:53
udp        UNCONN     0            0            192.168.1.229%enp0s3:68
udp        UNCONN     0            0            0.0.0.0:111
udp        UNCONN     0            0            127.0.0.1:161
udp        UNCONN     0            0            [::]:111
udp        UNCONN     0            0            [::1]:161
tcp        LISTEN     0            4096         127.0.0.54:53
tcp        LISTEN     0            4096         0.0.0.0:10050
tcp        LISTEN     0            4096         0.0.0.0:111
tcp        LISTEN     0            511         0.0.0.0:80
tcp        LISTEN     0            100         0.0.0.0:25
tcp        LISTEN     0            4096         0.0.0.0:2222
tcp        LISTEN     0            4096         127.0.0.53%lo:53
tcp        LISTEN     0            4096         [::]:10050
tcp        LISTEN     0            4096         [::]:111
tcp        LISTEN     0            100         [::]:25
tcp        LISTEN     0            4096         [::]:2222
lens@lens:~$ nmap IP_SERVEUR
Starting Nmap 7.94SYN ( https://nmap.org ) at 2026-03-29 22:21 UTC
Failed to resolve "IP_SERVEUR".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.10 seconds
lens@lens:~$ nmap IP_SERVEUR

```

Commandes a passer

```

lens@lens:~$ sudo ufw status
Status: active

To Action From
--
22 ALLOW Anywhere
80 ALLOW Anywhere
443 ALLOW Anywhere
2222/tcp ALLOW Anywhere
22 (v6) ALLOW Anywhere (v6)
80 (v6) ALLOW Anywhere (v6)
443 (v6) ALLOW Anywhere (v6)
2222/tcp (v6) ALLOW Anywhere (v6)

lens@lens:~$

```

```

Chain ufw-skip-to-policy-output (0 references)
target      prot opt source      destination
ACCEPT      all  --  anywhere    anywhere

Chain ufw-track-forward (1 references)
target      prot opt source      destination

Chain ufw-track-input (1 references)
target      prot opt source      destination

Chain ufw-track-output (1 references)
target      prot opt source      destination
ACCEPT      tcp  --  anywhere    anywhere    ctstate NEW
ACCEPT      udp  --  anywhere    anywhere    ctstate NEW

Chain ufw-user-forward (1 references)
target      prot opt source      destination

Chain ufw-user-input (1 references)
target      prot opt source      destination
ACCEPT      tcp  --  anywhere    anywhere    tcp dpt:ssh
ACCEPT      udp  --  anywhere    anywhere    udp dpt:22
ACCEPT      tcp  --  anywhere    anywhere    tcp dpt:http
ACCEPT      udp  --  anywhere    anywhere    udp dpt:80
ACCEPT      tcp  --  anywhere    anywhere    tcp dpt:https
ACCEPT      udp  --  anywhere    anywhere    udp dpt:https
ACCEPT      tcp  --  anywhere    anywhere    tcp dpt:2222

Chain ufw-user-limit (0 references)
target      prot opt source      destination
LOG         all  --  anywhere    anywhere    limit: avg 3/min burst 5 LOG level warn prefix "[UFW LIMIT
REJECT      all  --  anywhere    anywhere    reject-with icmp-port-unreachable

Chain ufw-user-limit-accept (0 references)
target      prot opt source      destination
ACCEPT      all  --  anywhere    anywhere

Chain ufw-user-logging-forward (0 references)
target      prot opt source      destination

Chain ufw-user-logging-input (0 references)
target      prot opt source      destination

Chain ufw-user-logging-output (0 references)
target      prot opt source      destination

Chain ufw-user-output (1 references)
target      prot opt source      destination
lens@lens:~$

```

```
Lens [Running] - Oracle VirtualBox: 1
File Machine View Input Devices Help

}
    limit rate 3/minute burst 10 packets counter packets 0 bytes 0 log prefix "[UFW ALLOW] "
}

chain ufw6-skip-to-policy-input {
    counter packets 0 bytes 0 drop
}

chain ufw6-skip-to-policy-output {
    counter packets 0 bytes 0 accept
}

chain ufw6-skip-to-policy-forward {
    counter packets 0 bytes 0 drop
}

chain ufw6-user-input {
    tcp dport 22 counter packets 0 bytes 0 accept
    udp dport 22 counter packets 0 bytes 0 accept
    tcp dport 80 counter packets 0 bytes 0 accept
    udp dport 80 counter packets 0 bytes 0 accept
    tcp dport 443 counter packets 0 bytes 0 accept
    udp dport 443 counter packets 0 bytes 0 accept
    tcp dport 2222 counter packets 0 bytes 0 accept
}

chain ufw6-user-output {
}

chain ufw6-user-forward {
}

chain ufw6-user-logging-input {
}

chain ufw6-user-logging-output {
}

chain ufw6-user-logging-forward {
}

chain ufw6-user-limit {
    limit rate 3/minute burst 5 packets counter packets 0 bytes 0 log prefix "[UFW LIMIT BLOCK] "
    counter packets 0 bytes 0 reject
}

chain ufw6-user-limit-accept {
    counter packets 0 bytes 0 accept
}
}
lens@lens:~$
```

```

}
lens@lens:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
  Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
  Active: active (running) since Sun 2026-03-29 22:00:17 UTC; 24min ago
  TriggeredBy: • ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
  Main PID: 869 (sshd)
    Tasks: 1 (limit: 2263)
  Memory: 2.1M (peak: 2.4M)
    CPU: 58ms
  CGroup: /system.slice/ssh.service
          └─869 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 29 22:00:16 lens systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 29 22:00:17 lens sshd[869]: Server listening on 0.0.0.0 port 2222.
Mar 29 22:00:17 lens sshd[869]: Server listening on :: port 2222.
Mar 29 22:00:17 lens systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
lens@lens:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:  sshd
lens@lens:~$

```

Questions

1- Différence entre firewall stateful et stateless?

Un firewall *stateful* analyse l'état des connexions et garde en mémoire le contexte, ce qui lui permet de distinguer un trafic légitime d'un trafic suspect. À l'inverse, un firewall *stateless* traite chaque paquet indépendamment, sans tenir compte de l'historique ou de l'état de la session.

2- Avantages de nftables par rapport à iptables.

Nftables offre une syntaxe plus simple et unifiée, remplaçant iptables, ip6tables, arptables et ebtables. Il est plus performant, plus flexible grâce aux sets et maps, et bénéficie d'une maintenance active contrairement à iptables qui est considéré comme obsolète.

3- Pourquoi toujours autoriser SSH avant d'activer un firewall?

Il est essentiel d'autoriser SSH avant de verrouiller un firewall, car sans cette règle tu risques de perdre l'accès distant à ton serveur et de ne plus pouvoir l'administrer.

4- Pourquoi désactiver la connexion root?

La désactivation de la connexion root renforce la sécurité, car root est une cible privilégiée des attaques par force brute. Utiliser un utilisateur normal avec sudo limite les risques en cas de compromission.

5- Avantages de changer le port SSH.

Changer le port SSH réduit les attaques automatisées, car la majorité des bots scannent le port 22 par défaut. Cela diminue le bruit dans les logs et complique légèrement la tâche des attaquants.

6- Rôle de Fail2Ban.

Fail2Ban surveille les journaux système et détecte les tentatives de connexion répétées ou suspectes. Lorsqu'une attaque est identifiée, il bloque automatiquement l'adresse IP fautive via le firewall pour une durée définie, protégeant ainsi le serveur contre les attaques par force brute.

Cocclusion.

Ce TD m'a permis de savoir comment analyser les ports ouverts, configurer un firewall simple, comprendre les chaînes de filtrage, . comparer différentes solutions de firewall Linux, analyser une configuration SSH, mettre en place l'authentification par clé et sécuriser les accès, donc la tache a été réussie.